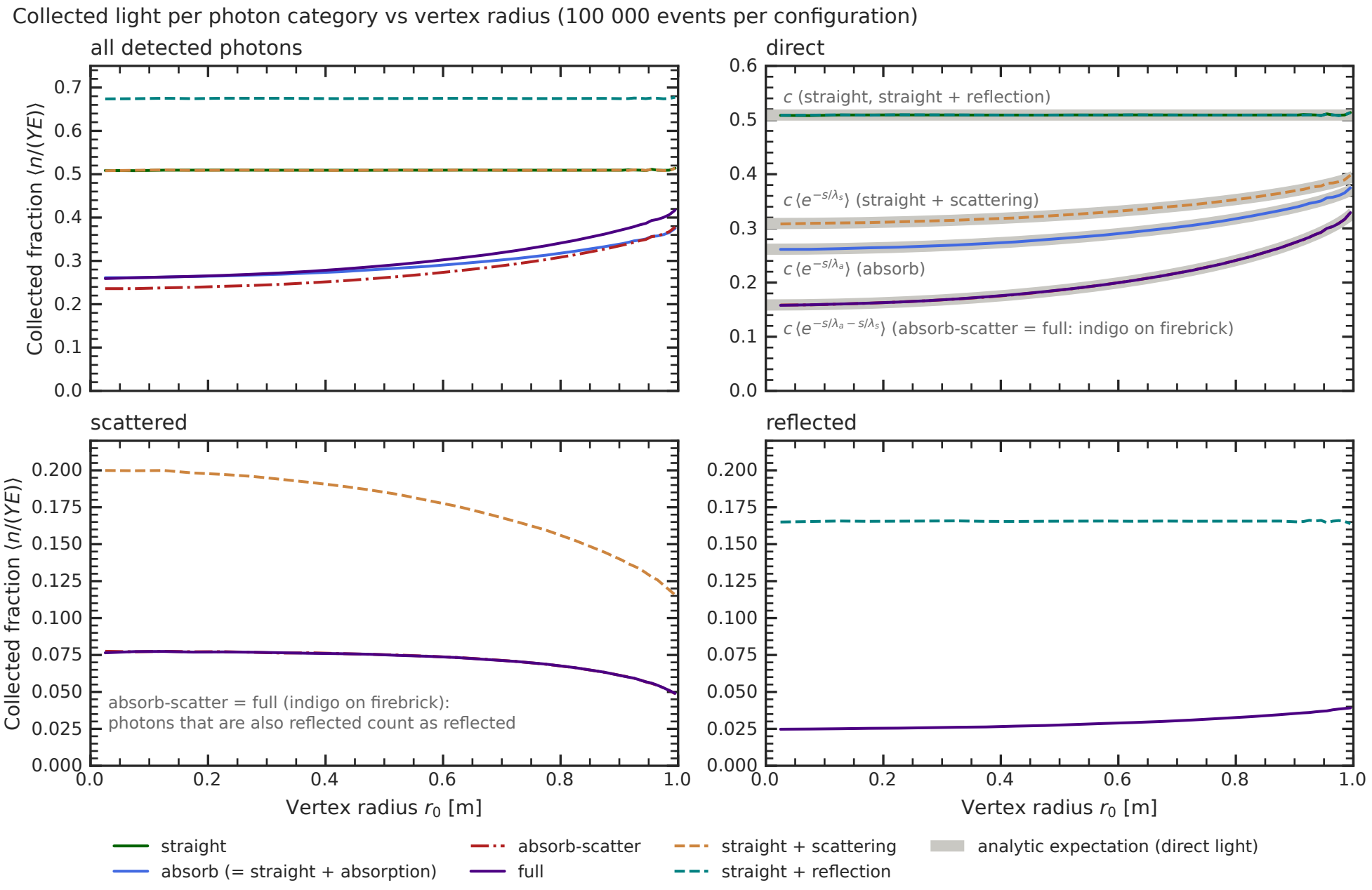
Collected light per photon category vs vertex radius (100 000 events per configuration)
all detected photons
direct
scattered
reflected
Collected fraction ⟨n/(YE)⟩
Vertex radius r_0 [m]
c (straight, straight + reflection)
$c\,\langle e^{-s/\lambda_s}\rangle$ (straight + scattering)
$c\,\langle e^{-s/\lambda_a}\rangle$ (absorb)
$c\,\langle e^{-s/\lambda_a - s/\lambda_s}\rangle$ (absorb-scatter = full: indigo on firebrick)
absorb-scatter = full (indigo on firebrick):
photons that are also reflected count as reflected
straight
absorb (= straight + absorption)
absorb-scatter
full
straight + scattering
straight + reflection
analytic expectation (direct light)